

FINARTS HW 3 — Market Model Regression

Group 4
Cuenca, Raphael
Galedo, Enrique Lorenzo
Go, Keira
Opiana, Aimee Lorynne A.
Patajo, Juliana
Sison, Aaron Joshua E.

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1 Overview

Model:

$$R_i = \beta_0 + \beta_1 \cdot R_m + u$$

Selected stocks (5 sectors):

Stock	RIC	Sector	Company
BPI	BPI.PS	Banks	Bank of the Philippine Islands
Century Pacific	CNPF.PS	Food Producers	Century Pacific Food
Meralco	MER.PS	Electricity	Manila Electric (Meralco)
ABS-CBN	ABS.PS	Media	ABS-CBN
PLDT	TEL.PS	Telecommunications	PLDT

2 Data Loading

```
# This code block loads required libraries, sets the random seed for reproducibility,
# and reads the raw Excel data files. It then filters for the 5 selected companies
# across 5 different sectors and displays their metadata.
library(dplyr)    # Used for data manipulation (filtering, mutating, summarizing)
library(tidyr)    # Used for reshaping data (pivoting from long to wide format)
library(readxl)   # Used for reading the raw Excel datasets into R
library(broom)    # Standard library for tidying regression outputs

set.seed(42L)

company_returns <- read_excel("data/raw/Company>Returns.xlsx")
psei_return     <- read_excel("data/raw/PSEi_Return.xlsx")
company_info    <- read_excel("data/raw/Company Name.xlsx")

selected_rics <- c("BPI.PS", "CNPF.PS", "MER.PS", "ABS.PS", "TEL.PS")

stock_meta <- company_info |>
  filter(RIC %in% selected_rics) |>
  select(RIC, Name, Sector, Full_Name = `Full Name`)

knitr::kable(stock_meta, caption = "Selected Stocks Metadata")
```

Table 2: Selected Stocks Metadata

RIC	Name	Sector	Full_Name
BPI.PS	BANK OF THE PHILP.ISLE.	Banks	Bank of the Philippines Islands
MER.PS	MANILA ELECTRIC	Electricity	Manila Electric
TEL.PS	PLDT	Telecommunications Service Providers	PLDT
CNPF.PS	CENTURY PACIFIC FOOD	Food Producers	Century Pacific Food
ABS.PS	ABS-CBN	Media	ABS-CBN

3 PART 1: Regression Analysis

Instruction: Select 5 stocks from 5 different sectors. Run a regression of the returns of each stock against the market returns.

$$R_i = \beta_0 + \beta_1 \cdot R_m + u$$

3.1 Question 1 (2 pts)

Create a dataframe for your regressions. It must contain the 5 stock returns, PSEi return, and date.

```
# This code block prepares the regression dataframe. It cleans the PSEi date column,
# joins it with the selected stock returns, and pivots the data into a wide format
# so each stock has its own column for daily returns.
psei <- psei_return |>
  mutate(Date = as.Date(Date)) |>
  rename(PSEi_Return = PSEi_Return) |>
  distinct(Date, .keep_all = TRUE)

stock_returns <- company_returns |>
  filter(RIC %in% selected_rics) |>
  rename(Return = Total_Return, Date = Date) |>
  mutate(Date = as.Date(Date))

reg_data_long <- stock_returns |>
  inner_join(psei, by = "Date")

reg_data_wide <- reg_data_long |>
```

```

select(Date, RIC, Return, PSEi_Return) |>
pivot_wider(
  id_cols      = c(Date, PSEi_Return),
  names_from   = RIC,
  values_from  = Return,
  names_prefix = ""
) |>
mutate(across(all_of(selected_rics), as.numeric)) |>
arrange(desc(Date))

knitr::kable(head(reg_data_wide, 5), digits = 4,
               caption = "Preview of the Regression Dataframe")

```

Table 3: Preview of the Regression Dataframe

Date	PSEi_Return	BPI.PS	MER.PS	TEL.PS	CNPF.PS	ABS.PS
2025-12-29	1.1823	-2.9264	-0.1739	0.0000	0.2571	-1.6355
2025-12-26	-1.4563	-0.1669	-0.8621	0.0000	-1.1436	4.3902
2025-12-23	0.4878	0.4191	-0.8547	0.0000	0.8974	-4.2056
2025-12-22	0.4902	2.8448	1.7391	0.6390	0.1284	13.8298
2025-12-19	0.4926	-2.2746	-2.8716	-1.4173	-2.1357	5.0279

3.2 Question 2 (2 pts each, 10 pts total)

Run the regressions and report your estimated parameters.

```

# This code block runs ordinary least squares regression for each selected stock
# against the PSEi market return. It extracts the intercept (B0) and slope (B1)
# along with their statistical significance and prints them in the requested format.
stock_names <- list(
  "BPI.PS" = "Bank of the Philippine Islands",
  "CNPF.PS" = "Century Pacific Food",
  "MER.PS" = "Manila Electric (Meralco)",
  "ABS.PS" = "ABS-CBN",
  "TEL.PS" = "PLDT"
)

stock_sectors <- list(
  "BPI.PS" = "Banks",
  "CNPF.PS" = "Food Producers",
  "MER.PS" = "Electricity",
  "ABS.PS" = "Media",
  "TEL.PS" = "Telecommunications Service Providers"
)

```

```

)

sig_stars <- function(p_value) {
  if (is.na(p_value)) return("")
  if (p_value < 0.01) return("***")
  if (p_value < 0.05) return("**")
  if (p_value < 0.10) return("*")
  return("")
}

regression_results <- list()

for (ric in selected_rics) {
  stock_data <- reg_data_long |> filter(RIC == ric)
  model <- lm(Return ~ PSEi_Return, data = stock_data)
  regression_results[[ric]] <- model

  coef_summary <- summary(model)$coefficients
  b0_est <- coef_summary[1, 1]; b0_pval <- coef_summary[1, 4]
  b1_est <- coef_summary[2, 1]; b1_pval <- coef_summary[2, 4]

  cat("STOCK", which(selected_rics == ric), "\n")
  cat("RIC:", ric, "\n")
  cat("Company Name:", stock_names[[ric]], "\n")
  cat("Sector:", stock_sectors[[ric]], "\n")
  cat("Estimated parameters:\n")
  cat("B0:", sprintf("%.6f", b0_est), sig_stars(b0_pval), "\n")
  cat("B1:", sprintf("%.6f", b1_est), sig_stars(b1_pval), "\n\n")
}

```

```

STOCK 1
RIC: BPI.PS
Company Name: Bank of the Philippine Islands
Sector: Banks
Estimated parameters:
B0: 0.062712
B1: 0.014535

```

```

STOCK 2
RIC: CNPF.PS
Company Name: Century Pacific Food
Sector: Food Producers
Estimated parameters:
B0: 0.087240
B1: -0.029359

```

```

STOCK 3

```

RIC: MER.PS
Company Name: Manila Electric (Meralco)
Sector: Electricity
Estimated parameters:
B0: 0.120955 *
B1: 0.033800

STOCK 4
RIC: ABS.PS
Company Name: ABS-CBN
Sector: Media
Estimated parameters:
B0: -0.025594
B1: 0.023995

STOCK 5
RIC: TEL.PS
Company Name: PLDT
Sector: Telecommunications Service Providers
Estimated parameters:
B0: 0.032580
B1: 0.065881 *

(Note on significance: ** $p < 0.01$, * $p < 0.05$, $p < 0.10$)*

3.3 Question 3 (5 pts)

Discuss your findings. Discuss the results for each stock, and compare the estimated parameters for each of the stocks.

The empirical estimation of the market model across the five selected equities reveals a pervasive lack of statistical significance at the standard 5% threshold ($p > 0.05$ for all β_1 coefficients). For the timeframe evaluated (2023–2025), the systematic market movements dictated by the PSEi failed to meaningfully explain the daily variance of these specific stocks, a finding corroborated by near-zero R^2 values across the panel. Consequently, the daily returns of these selected equities appear to be driven predominantly by idiosyncratic, company-specific factors rather than broader macroeconomic market trends.

Despite the overarching statistical insignificance, examining the individual slope coefficients provides insight into the directional sensitivities of the equities. PLDT (TEL.PS) registers the highest market sensitivity among the cohort with a beta of 0.0659, emerging as the only stock to approach marginal significance at the 10% level ($p = 0.083$). This suggests it tracks the broader index slightly more closely than its peers. Similarly, Manila Electric (MER.PS) yields a positive beta (0.0338) coupled with a marginally significant positive alpha ($\beta_0 = 0.1210$, $p < 0.10$), hinting at slight positive excess returns generated independently of the market. Both the Bank of the Philippine Islands

(BPI.PS) and ABS-CBN (ABS.PS) exhibit very weak positive betas (0.0145 and 0.0240, respectively) alongside high p-values, confirming that market fluctuations barely permeate their return profiles.

In contrast to the rest of the cohort, Century Pacific Food (CNP.F.PS) stands out as the only stock yielding a negative beta value. Specifically, the estimated parameters indicate that a 1% increase in the PSEi leads to a 0.029% decrease in Century Pacific's return. This inverse dynamic suggests a defensive, counter-cyclical characteristic inherent to the food producer, even though the relationship itself ultimately fails to achieve statistical significance ($p = 0.539$).

4 PART 2: Fitted Values and Residuals

4.1 Question 1 (5 pts)

Compute the fitted values of the 5 stocks that you analyzed.

$$\hat{R}_{i,t} = \hat{\beta}_0 + \hat{\beta}_1 \cdot R_{m,t}$$

```
# This code block computes the predicted (fitted) returns for all 5 stocks
# using the models estimated previously and adds them to the wide dataframe.
reg_data_wide <- reg_data_wide |>
  mutate(
    BPI_Fitted = fitted(regression_results[["BPI.PS"]]),
    CNPF_Fitted = fitted(regression_results[["CNP.F.PS"]]),
    MER_Fitted = fitted(regression_results[["MER.PS"]]),
    ABS_Fitted = fitted(regression_results[["ABS.PS"]]),
    TEL_Fitted = fitted(regression_results[["TEL.PS"]])
  )
```

4.2 Question 2 (5 pts)

Compute the residuals for each stock.

$$\hat{u}_{i,t} = R_{i,t} - \hat{R}_{i,t}$$

```
# This code block calculates the residuals for each stock by subtracting the
# fitted values from the actual returns. The result is stored as new columns.
reg_data_wide <- reg_data_wide |>
  mutate(
    BPI_Residual = BPI.PS - BPI_Fitted,
    CNPF_Residual = CNPF.PS - CNPF_Fitted,
    MER_Residual = MER.PS - MER_Fitted,
    ABS_Residual = ABS.PS - ABS_Fitted,
```

```
TEL_Residual = TEL.PS - TEL_Fitted
)
```

4.3 Question 3 (3 pts)

What did you notice about the sum of the residuals for each stock?

```
# This code block computes the sum of the residuals for each stock across all dates
# to demonstrate that the OLS residuals sum to approximately zero.
residual_sums <- reg_data_wide |>
  summarise(
    BPI_Sum = sum(BPI_Residual, na.rm = TRUE),
    CNPF_Sum = sum(CNPF_Residual, na.rm = TRUE),
    MER_Sum = sum(MER_Residual, na.rm = TRUE),
    ABS_Sum = sum(ABS_Residual, na.rm = TRUE),
    TEL_Sum = sum(TEL_Residual, na.rm = TRUE)
  )

knitr::kable(t(residual_sums), col.names = c("Sum of Residuals"),
  digits = 14, caption = "Sum of Residuals by Stock")
```

Table 4: Sum of Residuals by Stock

	Sum of Residuals
BPI_Sum	0.000000000000002
CNPF_Sum	-0.000000000000002
MER_Sum	0.000000000000003
ABS_Sum	0.000000000000006
TEL_Sum	0.000000000000005

The sum of residuals for every stock is practically zero. This is a fundamental mathematical property of Ordinary Least Squares regression when an intercept is included. The positive and negative errors perfectly cancel each other out, leaving only a microscopic remainder due to computer precision limits. In finance, these residuals represent the specific returns of a company that are independent of the broader market. A zero sum confirms that our market model is unbiased over the evaluation period. The model intercept fully captures the average excess return of the stock. Therefore, any firm specific surprises naturally average out to zero over time. This proves that the remaining unexplained returns represent purely diversifiable risk.

5 Data Export


```
# This code block exports the final combined dataset containing stock returns,  
# market returns, fitted values, and residuals into a CSV file.  
output_csv <- "H3_Group4_Regression_Output.csv"  
write.csv(reg_data_wide, file = output_csv, row.names = FALSE)
```

The final dataframe has been successfully exported to `H3_Group4_Regression_Output.csv` containing **727 rows** and **17 columns** (Date, PSEi_Return, 5 stock returns, 5 fitted values, 5 residuals).